AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

Chemistry Paper II

Time: 1 hour 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

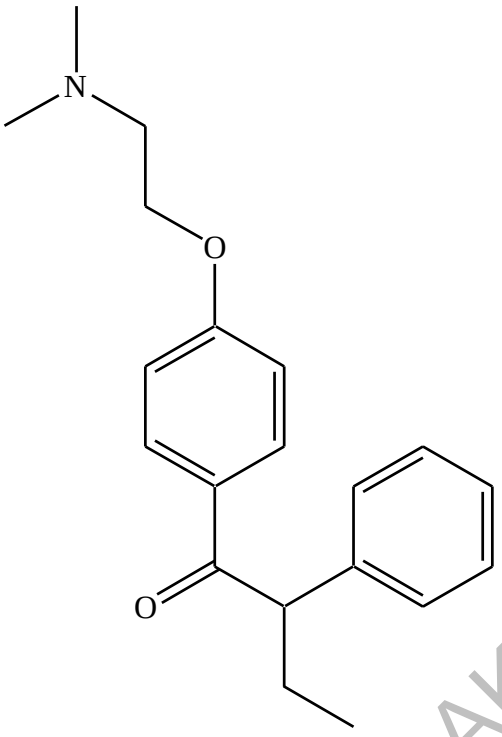
   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

Identify any THREE functional groups in the given structure of an organic molecule.



1. _____

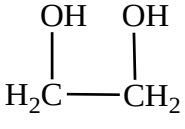
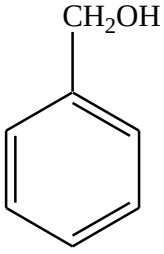
2. _____

3. _____

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Q.2. (Total 4 Marks)

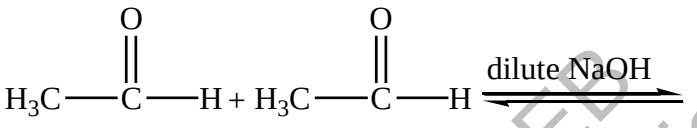
a. Write the names of the following alcohols as suggested by the International Union of Pure and Applied Chemistry (IUPAC). (2 Marks)

	Alcohol	IUPAC Name
i.		
ii.		

b. Briefly describe why the boiling points of alcohols are much higher than those of the corresponding alkanes. (2 Marks)

Q.3. (Total 4 Marks)

Consider the given chemical reaction.



a. Identify the type of reaction shown above. (1 Mark)

b. What will be the product of the given reaction? (1 Mark)

c. What will happen if the product of the given reaction is heated in the presence of dilute hydrochloric acid? Write a chemical equation for the reaction to support your answer. (2 Marks)

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Q.4. (Total 3 Marks)

Describe the role of the following types of RNA in protein synthesis in a human cell.

a. Messenger RNA (1 Mark)

_____

_____

b. Transfer RNA (1 Mark)

_____

_____

c. Ribosomal RNA (1 Mark)

_____

_____

Q.5. (Total 3 Marks)

In Pakistan, chemical industries are considered to have enough resources to manufacture chemicals independently; however, they prefer to import chemicals from foreign countries.

Write any THREE factors that cause Pakistani industries to import these chemicals.

_____

_____

_____

_____

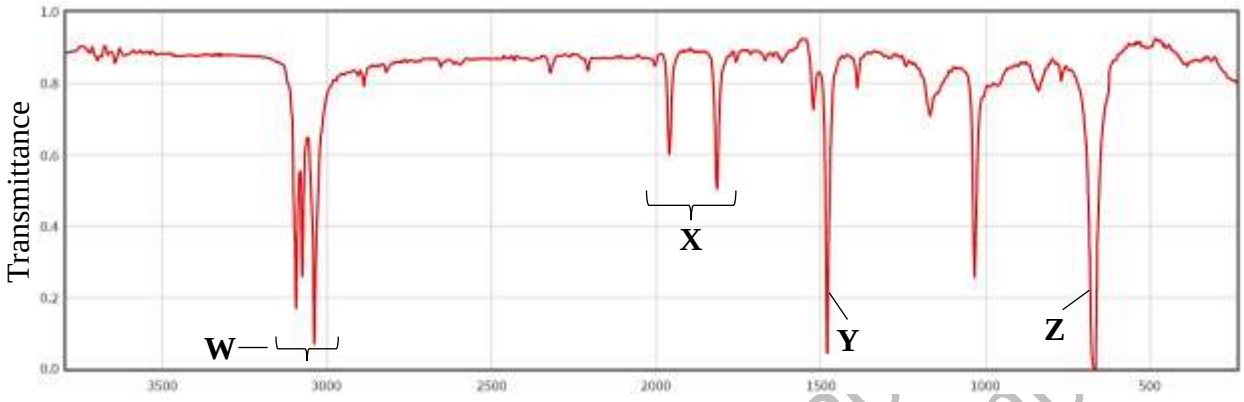
_____

_____

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Q.6. (Total 4 Marks)

Consider the given Infrared (IR) spectrum of an organic compound.



Wavenumber (cm^{-1})

Source: https://webbook.nist.gov/cgi/inchi?ID=C71432&Type=IR-SPEC&Index=2

a. Interpret any THREE labelled peaks (**W**, **X**, **Y** or **Z**) in the given spectrum. (3 Marks)

b. Based on the interpretation in part a, identify the compound. (1 Mark)

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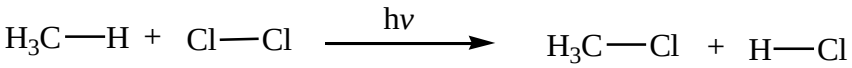
Q.7. (Total 7 Marks)

EITHER

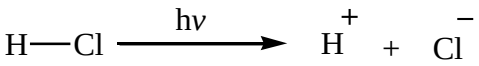
a.

i. Give TWO reasons why alkanes are mostly unreactive towards polar reagents. (2 Marks)

ii. Write the steps of the free radical mechanism by showing the initiation, propagation and termination steps for the given reaction. (4 Marks)



iii. Identify the type of fission reaction in the given chemical equation. (1 Mark)



OR

b.

i. Describe the importance of carbon dioxide gas in the atmosphere. (2 Marks)

ii. Write any FIVE harmful effects of the rise in concentration of carbon dioxide on the environment. (5 Marks)

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Q.8. (Total 7 Marks)

EITHER

a.

i. Write the order of bond strength in carbon halides, starting from the strongest to the weakest. (1 Mark)

ii. Based on the order of bond strength mentioned in part i, describe the following: (6 Marks)

I. Reactivity of alkyl halides

II. Solubility of alkyl halides in water and organic solvents

OR

b.

i. Identify which of the following readily undergoes nucleophilic substitution reaction. (1 Mark)

I. H_3CCOOH

II. $H_3CCOOCH_3$

ii. Describe reasons for each of the following statements.

I. Amides are more stable than esters. (2 Marks)

II. Hydrolysis of acyl halides occurs faster than anhydrides. (2 Marks)

III. Acyl chloride is the most reactive among the derivatives of carboxylic acid. (2 Marks)

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